Optimization

2.3 Successive quadratic interpolation (Powell's method)

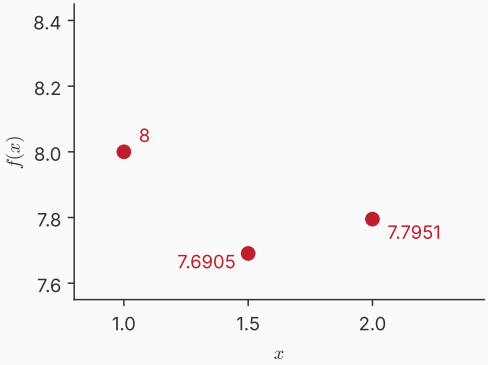
José Firmino Aguilar Madeira

The order or the numbers?

Three evaluations of the guiding example: $f(1) = 8$, $f(1.5) = 7.6905$, $f(2) = 7.7951$. Golden section would only use the order: $f(1.5)$ is the smallest, so $x^* \in [1, 2]$.

But the numbers say more: from 1 to 1.5, f drops by 0.31; from 1.5 to 2 it rises by only 0.10 — f is curving upwards, and the minimum should be to the right of 1.5.

Can we use the numbers to guess the shape of f?



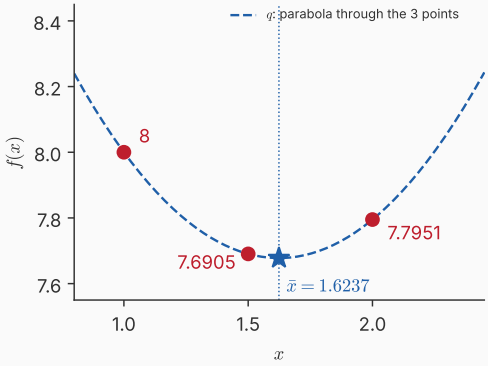
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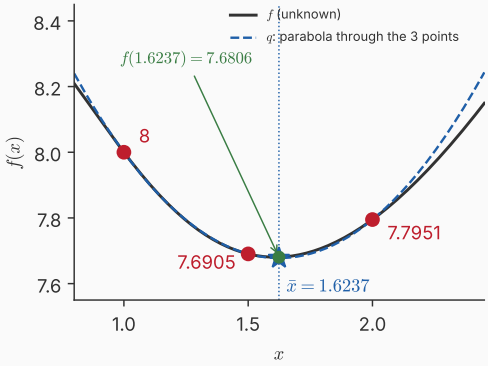
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Through the three points we pass the simplest model that curves: a parabola q.

If q were f, the minimum would be at $\bar{x} = 1.6237$. One new evaluation: $f(1.6237) = 7.6806$, 0.014 away from the true minimum ($x^* = 1.6094$).



If the parabola were f, where would the minimum be?

Parabola $q(x) = c_0 + c_1 x + c_2 x^2$ with $q(1) = 8$, $q(1.5) = 7.69050$, $q(2) = 7.79507$:
$q(x) = 0.82814x^2 - 2.68935x + 9.86121$ (check: $q(1) = 8$; $q(1.5) = 7.69050$; $q(2) = 7.79507$).
Minimum of the model: $q'(x) = c_1 + 2c_2 x = 0$,

$$\boxed{\bar{x} = -\frac{c_1}{2c_2}} = \frac{2.68935}{1.65628} = 1.6237.$$

A minimum because $c_2 = 0.828 > 0$ (with $c_2 < 0$ it would be a *maximum*).

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$q(1.6237) = 7.6778$ but $f(1.6237) = 7.6806$: the model is *not* f, but it points to the right region.

We keep the best point, 1.6237, and the two that bracket it, 1.5 and 2; fit a new parabola, and repeat. If f were a strictly convex quadratic, $q = f$ and $\bar{x} = x^*$ in a single iteration; near the minimum, a smooth function *looks like* a parabola.

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order → magnitude → model → prediction → new evaluation

This is the idea of successive quadratic interpolation (Powell's method).

Here, "Powell's method" is this *one-dimensional* interpolation; not to be confused with Powell's *multidimensional* derivative-free conjugate-direction method (e.g. `method='Powell'` in SciPy).

Computing $\bar{x}$ without solving a system

With any three points $x_1 < x_2 < x_3$, q is written in Newton form — the coefficients come out one at a time:

$$q(x) = a_0 + a_1(x - x_1) + a_2(x - x_1)(x - x_2),$$

$$a_0 = f_1, \qquad a_1 = \frac{f_2 - f_1}{x_2 - x_1}, \qquad a_2 = \frac{1}{x_3 - x_2}\left(\frac{f_3 - f_1}{x_3 - x_1} - a_1\right).$$

- a_1: slope of the chord between the first two points (first-order divided difference).
- a_2: second-order divided difference — it measures the curvature ($a_2 = f''(\xi)/2$ for some ξ); it is the coefficient of x^2, that is, $a_2 = c_2$. $a_2 > 0 \iff q$ has a minimum.

From $q'(\bar{x}) = a_1 + a_2(2\bar{x} - x_1 - x_2) = 0$:

$$\boxed{\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}}$$

In the example: $a_1 = -0.619$, $a_2 = 0.828$, $\bar{x} = 1.25 + 0.374 = 1.624$ — the same.

Algorithm: successive quadratic interpolation

Powell: x_1, step Δ, ε_x, ε_f

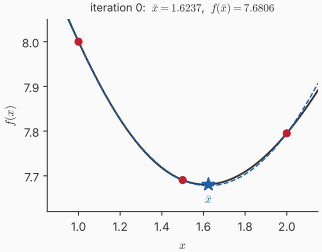
1: set $x_2 = x_1 + \Delta$; evaluate f_1, f_2
2: **if** $f_1 > f_2$: set $x_3 = x_1 + 2\Delta$; **else** set $x_3 = x_1 - \Delta$; evaluate f_3
3: **loop**
4: sort the three points: $x_1 < x_2 < x_3$
5: set $F_{\min} = \min(f_1, f_2, f_3)$ and $X_{\min}$ to the corresponding x_i
6: compute a_1, a_2; **if** $a_2 \le 0$: reflect the worst through the best; repeat
7: set $\bar{x} = \dfrac{x_1 + x_2}{2} - \dfrac{a_1}{2a_2}$; evaluate $f(\bar{x})$
8: **if** $|F_{\min} - f(\bar{x})| \le \varepsilon_f$ **and** $|X_{\min} - \bar{x}| \le \varepsilon_x$ **then**
9: **return** the best of the four points
10: **end if**
11: keep the best point and the two that bracket it; if they do not exist, the three best
12: **end loop**

- Steps 1–2: the third point is placed on the side where f decreases, to try to *bracket* the minimum from the start.
- If x_2 is the best and is *bracketed* by two worse points, the parabola is convex and its minimum lies inside (x_1, x_3).
- Cost: 3 initial evaluations + 1 per iteration (+1 for each reflection).
- Fast local convergence, without derivatives: superlinear (order ≈ 1.3) with the last three points; with bracketing it may be only linear.

Example: $f(x) = 8e^{1-x} + 7\ln x$, $x_1 = 1$, $\Delta = 0.5$

k	x_1	x_2	x_3	f_1	f_2	f_3	a_1	a_2	$\bar{x}$	$f(\bar{x})$
0	1.0000	1.5000	2.0000	8.0000	7.6905	7.7951	−0.619	0.828	1.6237	7.6806

Start: $x_2 = 1.5$; $f(1) = 8 > f(1.5) = 7.6905 \Rightarrow x_3 = 1 + 2\Delta = 2$, $f(2) = 7.7951$.

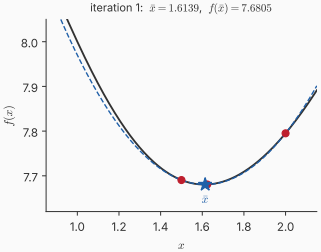


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After it. 0: the best is $\bar{x} = 1.6237$, bracketed by 1.5 and 2: we keep 1.5, 1.6237, 2. The point 1.5 stays to the end: all the $\bar{x}$ fall to the right of x^*.

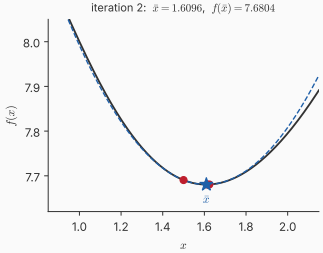


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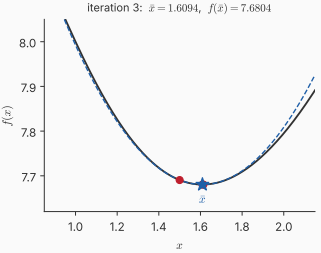
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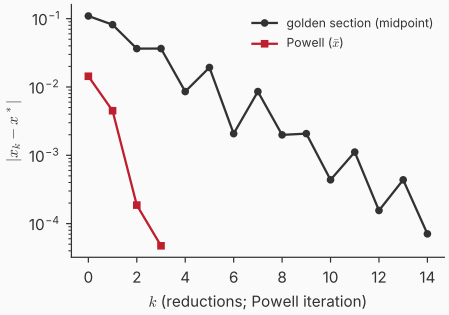
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3	1.5000	1.6096	1.6139	7.6905	7.6804	7.6805	−0.092	0.840	1.6094	7.6804

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After it. 0: the best is $\bar{x} = 1.6237$, bracketed by 1.5 and 2: we keep 1.5, 1.6237, 2. The point 1.5 stays to the end: all the $\bar{x}$ fall to the right of x^*.
End (it. 3, the 4th): $|F_{\min} - f(\bar{x})| < 10^{-3}$ and $|X_{\min} - \bar{x}| < 10^{-3}$.
$x^* \approx 1.6094$, $f(x^*) \approx 7.6804$; 7 evaluations of f. (2019 first test: it. 0 and 1.)



Golden section vs. Powell on the same problem



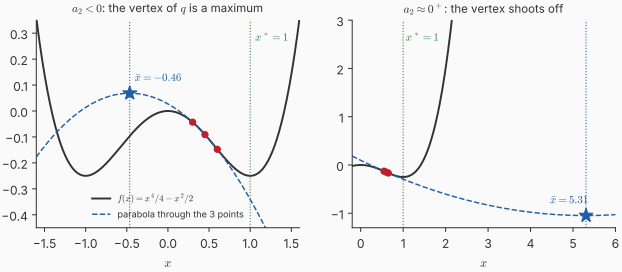
- Golden section on $[1, 2]$: interval $\times 0.618$ per reduction; the error of the midpoint oscillates and falls below 10^{-3} after 12 reductions (guarantee $L_k/2$: 13).
- Powell: error $0.014 \to 0.0045 \to 0.0002 \to 5 \cdot 10^{-5}$ — 4 iterations, 7 evaluations.

But

With f unimodal, golden section always converges; quadratic interpolation may give $a_2 \leq 0$ or jump far away (next page). In practice they are combined (`fminbnd`).

Note — finite precision (bridge to 2.6): no method that only compares values of f can distinguish differences below $\varepsilon_m |f^*|$; here this limits $|x - x^*|$ to about $\sqrt{2\varepsilon_m |f^*| / f''(x^*)} \approx 5 \cdot 10^{-8}$.

When the parabola does not work



- $a_2 < 0$ (left, points in a concave region): q has a *maximum*; $\bar{x} = -0.46$ goes the wrong way.
- $a_2 \approx 0^+$ (right, points almost aligned near an inflection): the vertex *shoots off* to $\bar{x} = 5.31$, $f(\bar{x}) = 184$.
- Remedy: if $a_2 \leq 0$, do not compute $\bar{x}$ — reflect the worst point through the best (moving in the direction in which f decreases); and, as soon as possible, keep the best point bracketed between two worse ones: then $a_2 > 0$ and $\bar{x}$ falls inside the interval. The case of small $a_2 > 0$ is solved only by bracketing; `fminbnd` (Brent) also limits the step.

On the computer

The core of one iteration, in MATLAB:

```
[X, i] = sort(X);  F = F(i);          % x1 < x2 < x3
a1 = (F(2)-F(1)) / (X(2)-X(1));
a2 = ((F(3)-F(1))/(X(3)-X(1)) - a1) / (X(3)-X(2));
if a2 <= 0           % parabola with no minimum:
  % reflect the worst point through the best and repeat
else
  xb = (X(1)+X(2))/2 - a1/(2*a2);
  fb = f(xb);                         % 1 evaluation
  % stop if xb and fb change little; otherwise keep
  % the best point and the two that bracket it
end
```

The full function (`PowellQuadratic`, with the safeguards and `info.nfev`) is in the course companion code.

In the example ($x_1 = 1$, $\Delta = 0.5$, tolerances 10^{-3}): 4 iterations, no reflections: $n_f = 3 + 4 = 7$.

Safeguard: with $a_2 \leq 0$ the best point is at an end. Reflecting the worst through the best, $2X_{\min} - x_{\text{worst}}$, generates a point on the other side of the best, outside the trio, in the direction in which the observed values indicate descent. In class: the three best and, with $a_2 \leq 0$, a random point (Deb's suggestion); here, bracket (Deb, Step 7) and reflect (deterministic). Same result in the exercises.

MATLAB's `fminbnd` (Brent) alternates between parabolic steps and golden section steps: the speed of one, the guarantee of the other.

KEY TAKEAWAYS — 2.3

1. Golden section uses only the *order* of the values of f; successive quadratic interpolation also uses their magnitude: it fits a parabola to three points and evaluates f at its minimum.
2. $\bar{x} = \dfrac{x_1 + x_2}{2} - \dfrac{a_1}{2a_2}$, with a_2 (curvature) > 0. On a strictly convex quadratic, a single iteration.
3. Model $\rightarrow$ prediction $\rightarrow$ new evaluation: one evaluation per iteration and, near the minimum, locally and under suitable assumptions, fast convergence (superlinear, order ≈ 1.3, in the variant with the last three points).
4. On its own it has no guarantee: $a_2 \leq 0$ or almost aligned points require safeguards (reflection, bracketing); in practice it is combined with golden section (`fminbnd`).

Next (2.4): what if we have f'? At an interior minimum $f'(x^*) = 0$. As in order 0, we start by comparing: the *sign* of f' tells on which side the zero is — bisection.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987 — line search by interpolation.

Resources

- Companion code: `PowellQuadratic` (safeguards, `info.nfev`). MATLAB: `fminbnd` (Brent).